

Qatar-1b

R-0320

Benchmark run · workflow W8-benchmark-deep · record deep-bench_Qatar-1_170516 · created 2026-07-18T01:58:40+00:00

BENCHMARK: INACCURATE

Results

Mid-Transit Time (Tmid)	2457889.85342 +/- 0.00092 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.192 +/- 0.012
Transit depth (Rp/Rs)^2	3.7 +/- 0.45 [%]
Orbital Inclination (inc)	83.96 +/- 1.38
Airmass coefficient 1 (a1)	1.015 +/- 0.019
Airmass coefficient 2 (a2)	-0.0125 +/- 0.0082
Scatter in the residuals of the lightcurve fit is	1.83 %
Best Comparison Star	#3 - [484, 145]
Optimal Aperture	3.28
Optimal Annulus	12.65
Transit Duration (day)	0.0709 +/- 0.008

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.192 $\pm$ 0.012 vs 0.1463 (deviation 3.81 σ)
Duration fit vs published	0.0709 d vs 0.0692 d (deviation 0.22 σ)
Mid-time O-C	2.6 min
Depth SNR	16.0
Residual scatter	1.83 %

Light curve

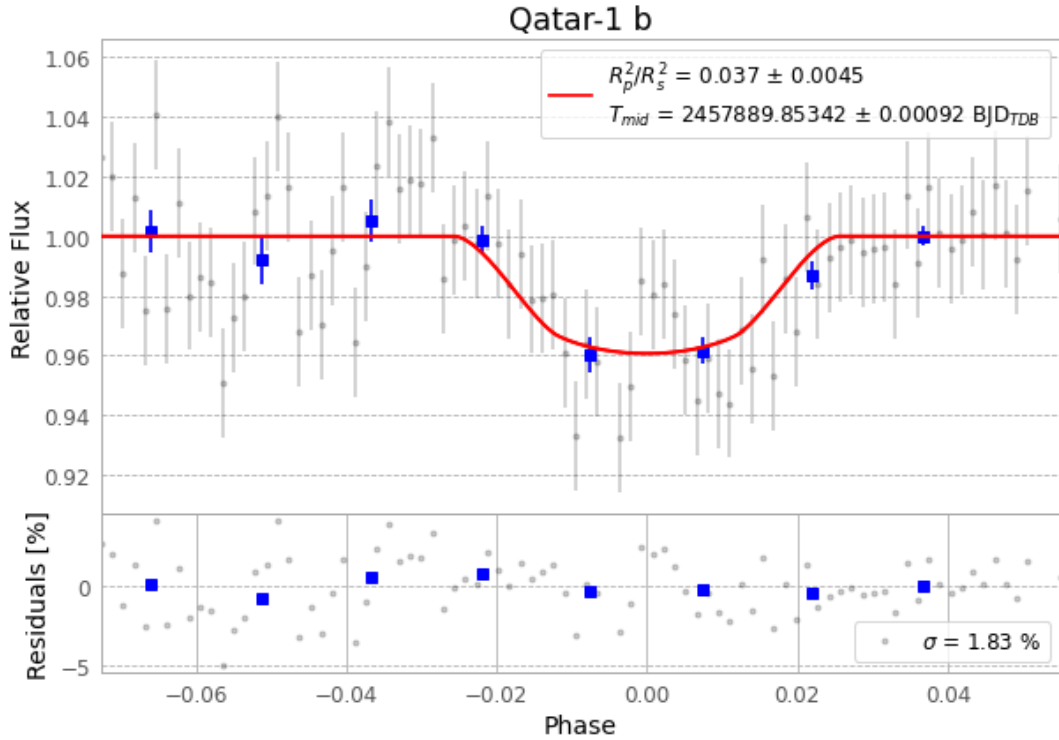


Figure 1 — FinalLightCurve_Qatar-1 b_2017-05-15.png (SHA-256 fc4d2ba12336a678...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [177, 214], 3 comparison star(s). Exact configuration: parameters.inits.json in record.json (SHA-256 c750bcf680586bab...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit 1bbef5c

Data provenance

101 FITS frames (66 MB), Qatar-1170516060012.FITS → Qatar-1170516110012.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-Qatar-1-170516.log (fcfce12d1bbb...), FinalLightCurve_Qatar-1 b_2017-05-15.png (fc4d2ba12336...), FinalLightCurve_Qatar-1 b_2017-05-15.csv (5944c6bbe452...)

Compute resources

Wall time 510.6 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-18 01:58Z.